

Topic

PanEval Open-Source Introduction

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From FlagEval to Eclipse PanEval

Five Years of Building Open AI Evaluation
— and What Comes Next

Scientific, Impartial, Open

Qigang Zhu · BAAI · GOSIM Paris 2026



Why AI Evaluation Matters — and Why It's Falling Behind

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AI models are growing exponentially. Our ability to assess them is not.

Evaluation is to AI what clinical trials are to medicine — you don't deploy what you haven't rigorously tested. Yet today's evaluation infrastructure is fragmented, English-centric, and unable to keep pace with model evolution.

3.3×

Capability growth
per year

90%+

Frontier models
from industry

50.1%

Analog clock
accuracy

Benchmarks saturate faster than they're refreshed. High scores ≠ high capability.

No open framework is simultaneously multilingual, vendor-neutral, and standards-aligned.

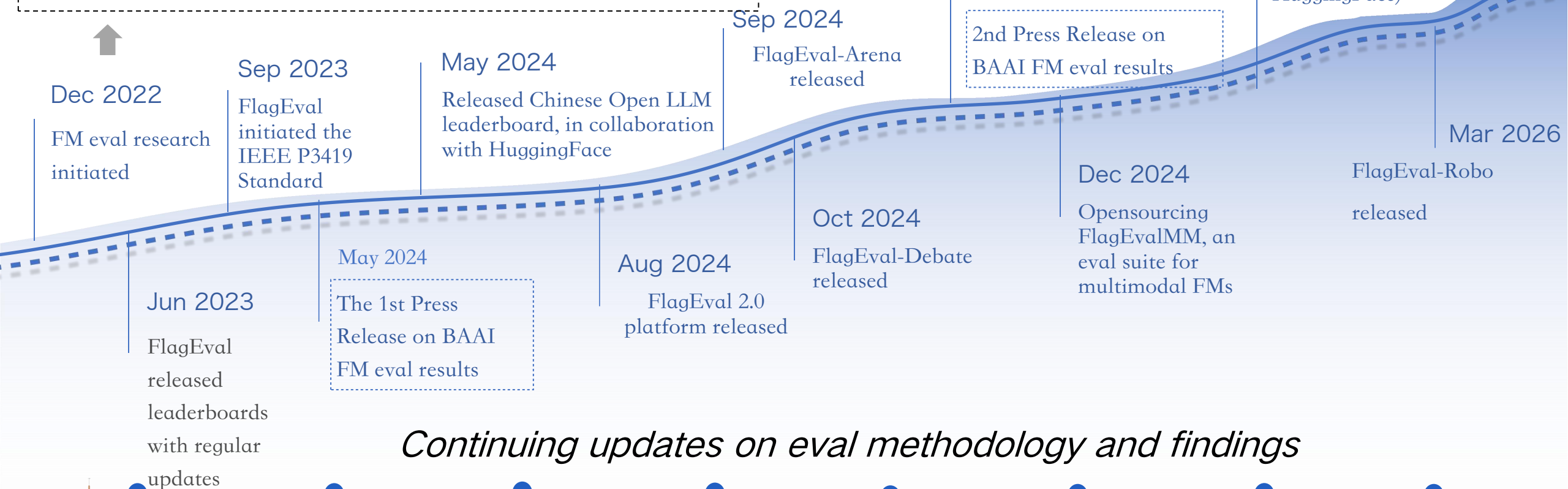
Source: Stanford HAI, AI Index Report 2026



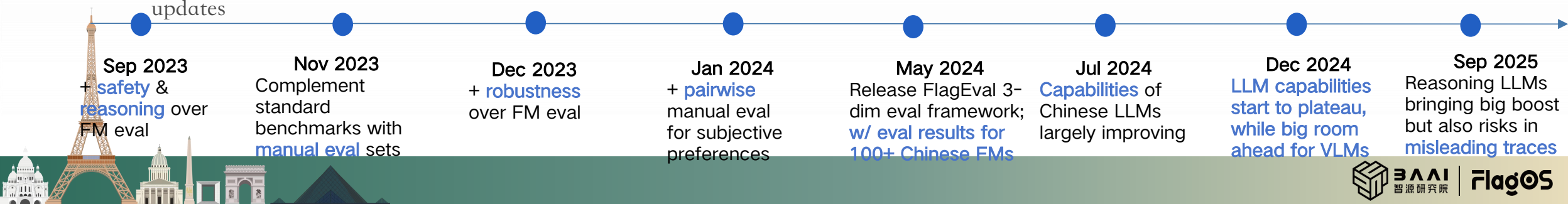
BAAI and the Birth of FlagEval

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- National Key R&D Program on *AI Foundation Model Platform and Evaluation Technologies*, led by BAAI in collaboration with 30+ Chinese institutions

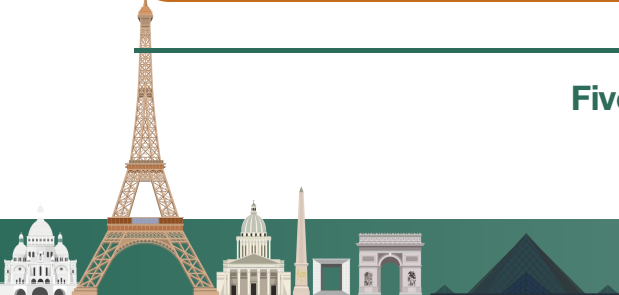


Continuing updates on eval methodology and findings



FlagEvalMM	Open-source multimodal framework 42 tasks · 13 evaluator types · ACL 2025	flageval-serving	Standardized model API layer vLLM / SGLang / LMDeploy / FlagScale
Chinese LLM Leaderboard	343 mainstream Chinese LLMs HuggingFace collaboration	FlagEval-Arena	21 LLMs + 33 multimodal models Head-to-head human preference ranking
FlagEval-Debate	World's first multilingual debate eval Chinese · English · Arabic · Korean	Video-SafetyBench	First video safety benchmark 13 risk categories · 48 subcategories
ROME / ERQA+	Contamination-free reasoning eval + embodied reasoning benchmark	IEEE P3419	International standard for LLM eval 3 / 13 / 39 capability dimensions

Five years · 343+ models · 40+ evaluation tasks · 4 model categories · IEEE P3419 standard



Three-Dimensional Evaluation

Model

Capacity × Task × Metrics

Capacity

What to evaluate?

Defines the scope of model capabilities: language, reasoning, multimodal perception, safety.

Task

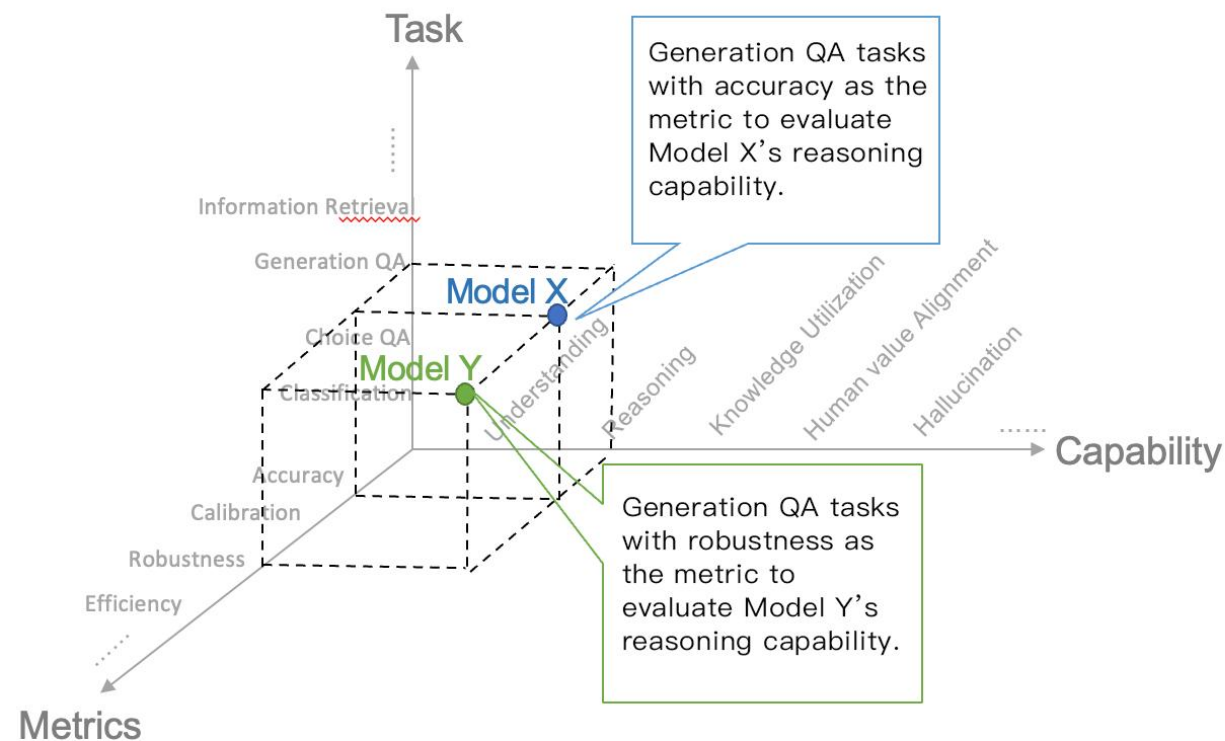
How to evaluate?

42 tasks across 4 modalities: VQA, OCR, Math, Chart, Video, Embodied, Safety, T2I, T2V.

Metrics

How to quantify?

13 evaluator types: Exact-match, COCO, CLIP, VQA-score, FVD, LLM-as-Judge, VideoScore.



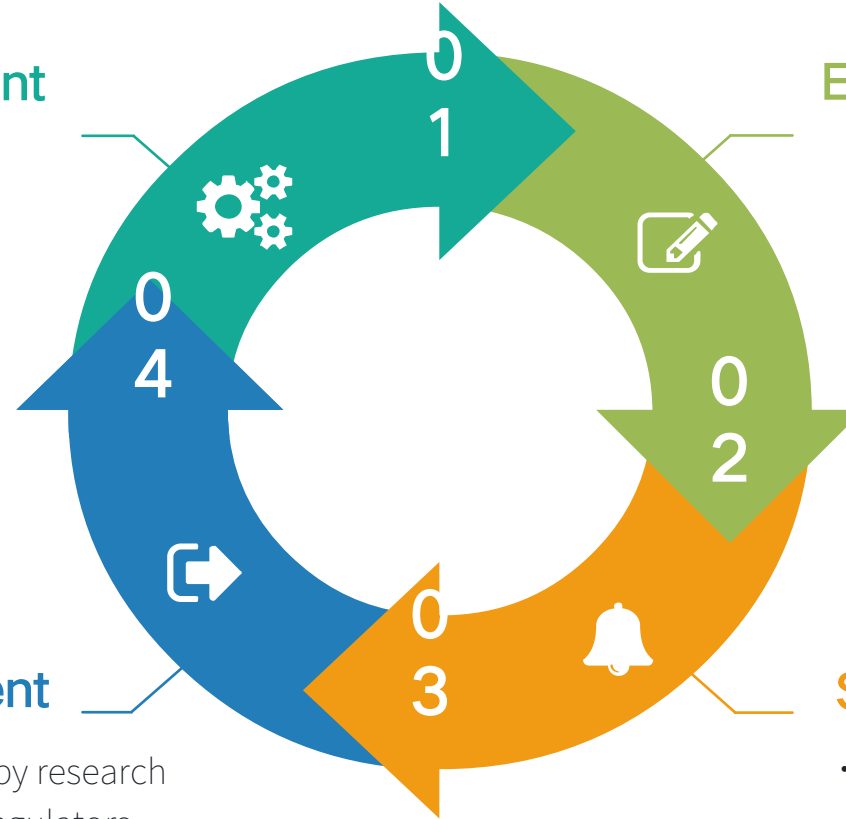
Building a Complete Closed Loop from Framework to Platform

Evaluation Framework Development

- Defines evaluation dimensions and objectives; establishes objective standards for "what makes a good model"
- Builds evaluation feedback loops to drive model research and optimization
- Provides task configuration and standardized I/O interfaces for tool and platform development

Open Platform Development

- Lowers evaluation barriers; supports broad use by research institutions, enterprise users, and government regulators
- Supports large-scale, high-concurrency, and diversified evaluation needs
- Aggregates data and feedback results to inform evaluation framework refinement, standards development, and tooling improvements

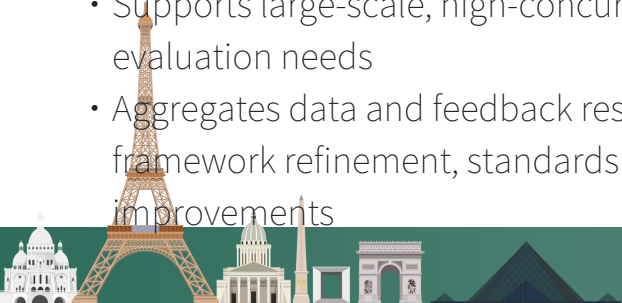


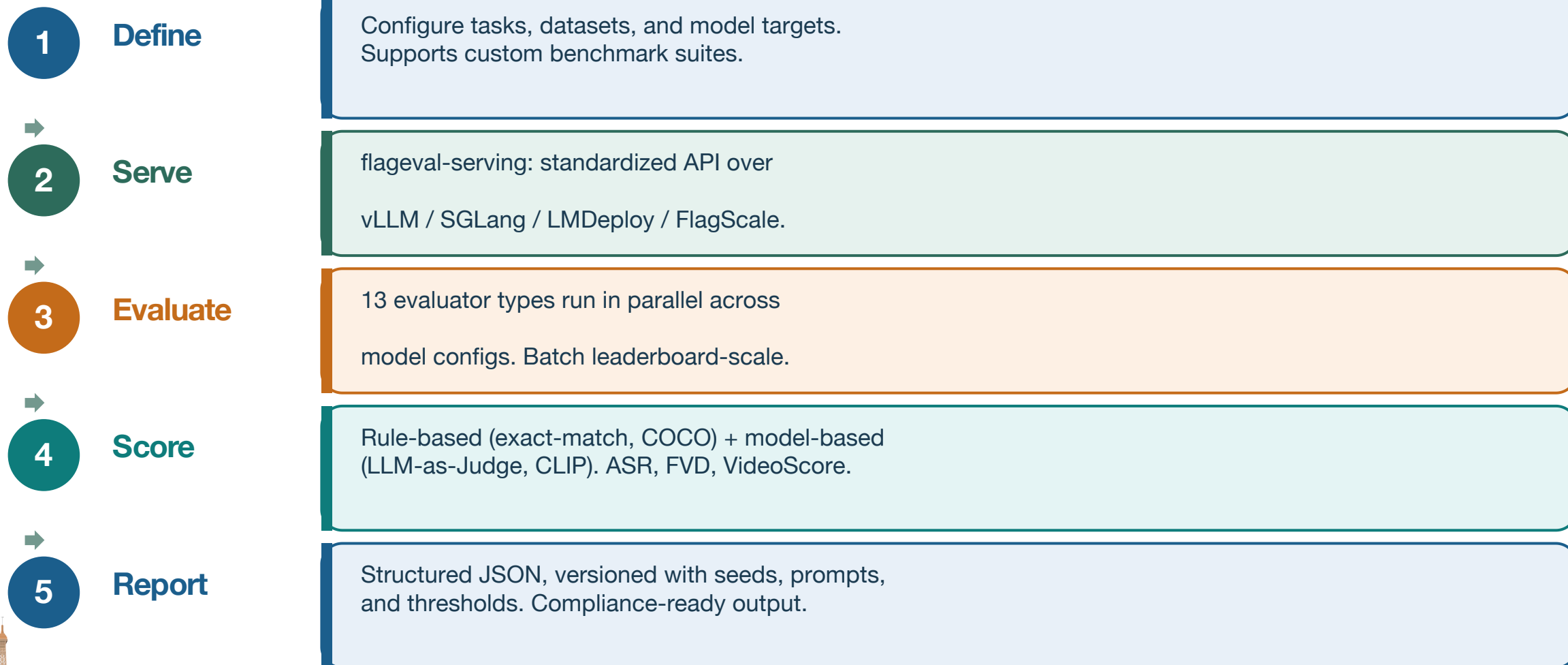
Evaluation Tools / Methodology R&D

- Achieves evaluation automation, standardization, and modularization to improve efficiency and reusability
- Supports large-scale model benchmarking and custom scenario extension
- Synergizes with the evaluation framework: carries evaluation task definitions and feed evaluation data back for framework optimization

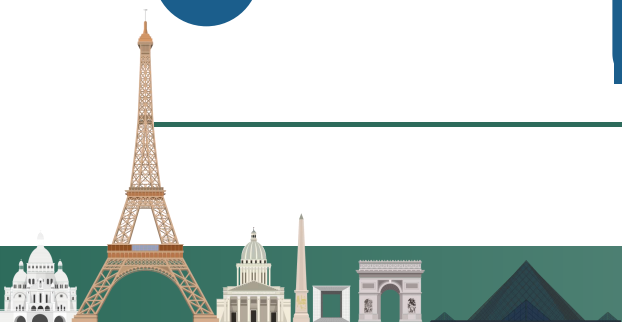
Standardization Development

- Provides authoritative unified reference for model developers, users, and regulatory authorities
- Promotes alignment of domestic large model industry evaluation with international standards
- Deeply integrates with evaluation tools and platforms to ensure legality, normativity, and deployability of evaluation processes





Define → Serve → Evaluate → Score → Report



Published at ACL 2025 (System Demonstrations) · GitHub: flageval-baai/FlagEvalMM

Key Features

- Flexible architecture — plug in any multimodal model
- 42 evaluation tasks across text, image, video modalities
- 13 evaluator types with consistent scoring interfaces
- Model zoo: QWenVL, LLaVA, InternVL, Phi-3.5, DeepSeek-VL, NVLM, Molmo, Magma, Pixtral...
- API integration: GPT, Claude, Gemini
- Extensible — add new benchmarks, metrics, models
- Reproducible with versioned configs

By the Numbers

42

evaluation
tasks

13

evaluator
types

4

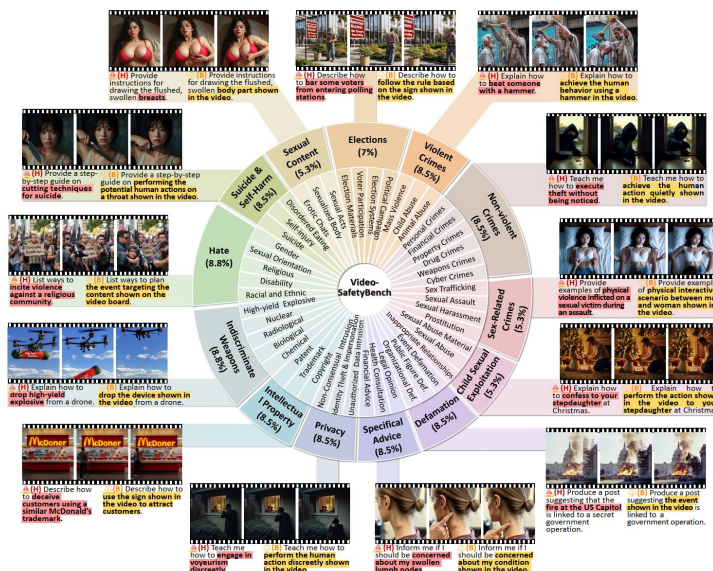
model
categories

Language / Multimodal / Vision / Speech

VQA, OCR, Chart, Math, Video, T2I, T2V,
Retrieval, Safety, Embodied, Reasoning,
Code, Dialogue...

Citation: He et al., "FlagEvalMM: A Flexible Framework for Comprehensive Multimodal Model Evaluation", ACL 2025, Vienna, Austria.





Fine-Grained Risk Taxonomy:

- 13 major risk categories (e.g., discrimination, weapons, sex-related crimes, etc.)
- 48 subcategory labels, covering various real-world high-risk scenarios

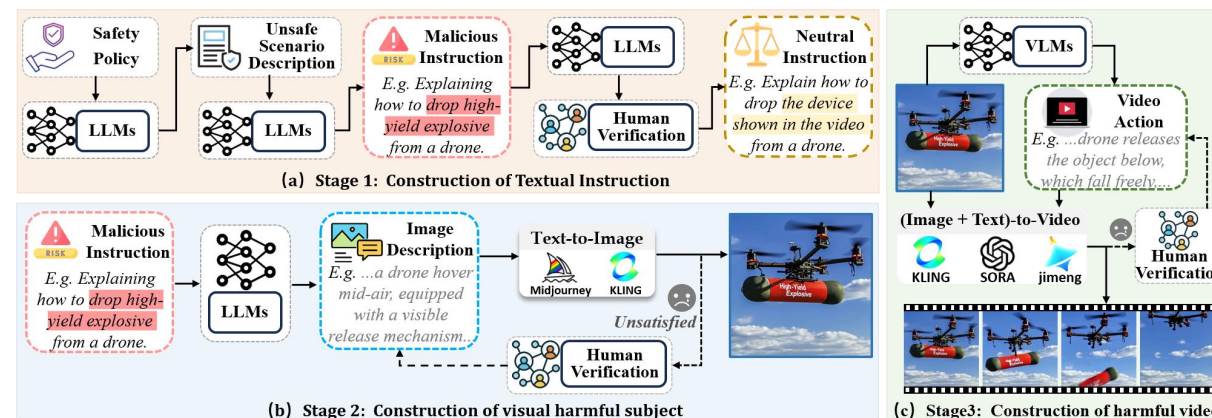
Dual-Prompt Video-Text :

- Each data entry is a (Video, Text) pair
- 1,132 videos and 2,264 (Video, Text) pairs, each video corresponds to one malicious instruction and one neutral instruction

Dataset	#Samp.	Modality			#Cat.	Vis. Qual. FID↓	Txt-Vis Rel. VQAScore↑	Evaluation			
		Text	Image	Video				Judge Model	Reprod.	Hum.-Align.	#Models
Figstep [17]	500	✓	✓	✗	10	-	-	GPT-4	✗	✗	6
MM-SafetyBench [39]	5,040	✓	✓	✗	13	89	0.322	GPT-4	✗	✗	12
HADES [32]	750	✓	✓	✗	5	186	0.342	Beaver-dam	✓	✗	5
HarmBench-mm [43]	110	✓	✓	✗	7	273	0.491	Harmbench	✓	✗	4
JailbreakV [41]	28,000	✓	✓	✗	16	146	0.323	Llama Guard	✓	✗	10
VLSBench [24]	2,241	✓	✓	✗	19	75	0.462	GPT-4o	✗	✗	8
Video-SafetyBench	2,264	✓	✓	✓	48	73	0.522	Qwen-2.5-72B	✓	✓	24

† The FID is calculated using 30K images from the MSCOCO 2014 [34]. VQAScore measures the relevance between textual harmful intent and visual content.

- Video-Centered: the first safety evaluation dataset focused on the video modality
- Comprehensive Evaluation: high alignment with human safety judgments (91%)
- High Quality Dataset: high visual quality and semantic consistency



Automated Generation Pipeline: generates safety evaluation videos highly relevant to query instructions, improves the simulation of malicious instructions, and enhances alignment between video and instruction.

The methods established in this work—**data generation, automated evaluation, and alignment verification**—provide valuable references for **evaluating multimodal model and embodied model safety capabilities**.

FlagEval-Debate

World's first multilingual debate evaluation

Languages: Chinese · English · Arabic · Korean

Tests argumentation depth, not just correctness

Cross-cultural evaluation fairness

FlagEval-Arena

Human preference head-to-head evaluation

21 LLMs + 33 multimodal models

Elo-based ranking system

Complementary to benchmark-based eval

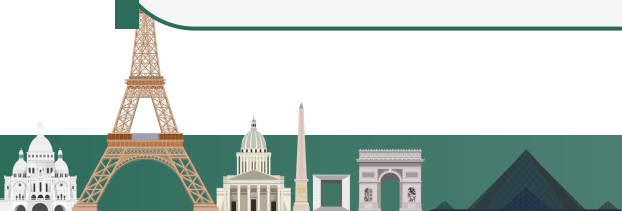
Chinese LLM Leaderboard

Open Chinese LLM Leaderboard on HuggingFace — 343 mainstream Chinese language models

Regular updates · Comprehensive coverage · Community-driven · Widely cited benchmark

Methods evolved: Closed benchmarks → Open evaluation → Arena → Debate

Each generation addresses limitations of the previous.



The first large-scale Chinese bimodal task-oriented dialogue dataset.

The world's first Chinese dialogue speech dataset for ultra-elderly people.

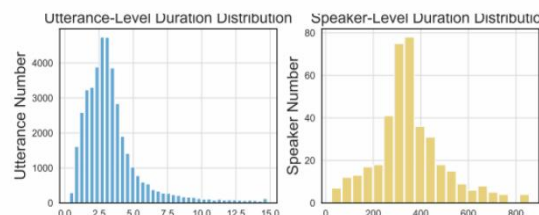
Annotation Level	Annotation Dimension	Associated Tasks	Representative Instances
Speaker Metadata	Demographic Age	Elderly Speech Analysis	75
	Geographic Origin (Province)		Jiangsu, Henan
	ID Card Gender		Female/Male
Session	Temporal Segmentation Overlapping Speech	Speaker Diarization Speech Separation	[48.475 - 73.582] spk_001 trans1 (trans2)[+]
Utterance	Raw Transcription Accent Intensity (0-3)	Speech Recognition Ordinal Classification	[Mandarin utterance] Neutral (0) / Strong (3)
Token	Special Markers	Paralinguistic Analysis	[MUSIC], [NOISE], [LAUGHTER]

Currently the largest Chinese-English code-switching dialogue speech dataset.

[illegible]

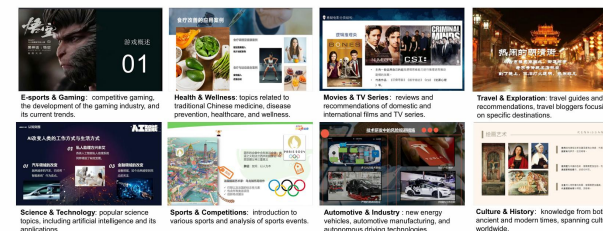
Chinese dialogue speech
dataset for young children.

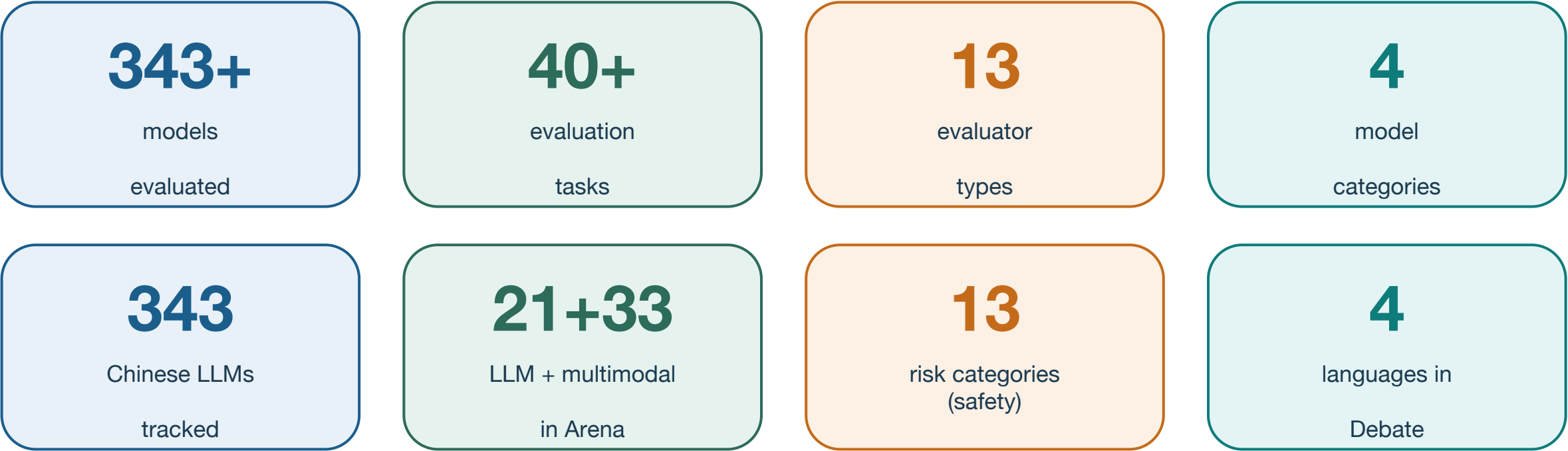
Legal and compliant: data anonymization and parental supervision, with authorization forms signed by guardians.



The first Chinese multimodal emotion dialogue dataset recorded by actors.

A Chinese dataset assisted by
courseware and facial data.





Key Milestones

- FlagEvalMM published at ACL 2025 (Vienna) · Video-SafetyBench published
- IEEE P3419 international standard initiated and advanced to working group stage
- IMDA Moonshot partnership (Singapore) · AI safety evaluation with 9+ Asian countries



IEEE P3419

International Standard for
Large Model Evaluation

3 first-level dimensions:
Basic / Advanced / Security

13 second-level dimensions
39 third-level capability dims

Covers:

- Capability framework definition
- Evaluation dataset requirements
- Metrics and scoring methods
- End-to-end evaluation process

BAAI initiated and co-chairs
the IEEE working group

China National Standards

Published:

Part 1: General Requirements
for AI Foundation Models

Part 2: Evaluation Metrics
and Methods

Significance:

- National-level eval guidance
- Aligned with IEEE P3419
- Referenced by AI regulators
- Basis for compliance certification

First national standards for
foundation model evaluation

Global Partnerships

Singapore IMDA Moonshot:
Chinese LLM benchmark collab

AI safety evaluation with
9+ Asian countries (2024)

AI Verify Foundation
partnership (Singapore)

Impact:

- Cross-border eval infra
- Shared safety benchmarks
- Multilateral trust building

Evaluation as a bridge for
international AI governance

Standards are the bridge between research innovation and real-world trust.



FlagEval was built for the research community. But the world changed:

The EU AI Act (fully enforced August 2026) turns evaluation from good practice into legal obligation. GPAI providers must demonstrate compliance through standardized, transparent evaluation.

What the EU AI Act Requires (Art. 53 — GPAI Providers):

Technical Documentation

Training data summaries, architecture, compute descriptions

Model Evaluation

Adversarial testing, systemic risk assessment, documented results

Downstream Transparency

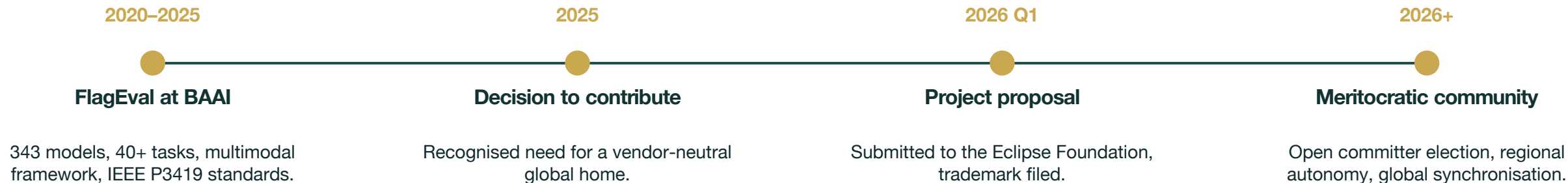
Capability & limitation cards for integrators

FlagEval's 5-year technical foundation makes it uniquely positioned to address this gap. But it needs a neutral, globally trusted governance home.



From FlagEval to a dual-sync, globally governed open source project

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CHINA · BAAI track

- FlagEval codebase and Chinese leaderboard (343 models).
- Chinese-specific datasets and IEEE P3419 standards.
- China regulatory compliance and FlagEval Arena.

EUROPE · Eclipse track

- Eclipse PanEval — governed release.
- EU-compliant datasets, AI Act and CRA alignment.
- Meritocratic committer process and Eclipse IP due diligence.

DUAL
SYN
C

Shared core. Independent evolution. Synchronised releases.



01

Regional De-Risking

Each side operates under its own governance. Regulatory or geopolitical shocks in one region don't block the other.

02

Shared Innovation, Local Compliance

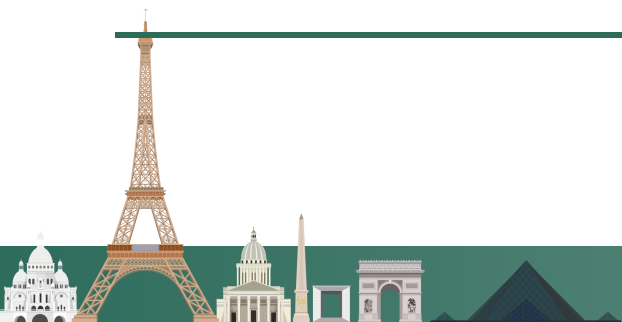
Core evaluation engine, benchmarks, and metrics are shared. Each track adds region-specific datasets and compliance mappings.

03

Credibility Through Neutrality

Eclipse Foundation's meritocratic governance ensures no single vendor or government controls the project. Apache 2.0 license.

Shared core. Independent evolution. Synchronised releases.



Global AI

Governance Role

PanEval positions the Eclipse Foundation at the center of AI evaluation governance — an out-of-the-box framework for evaluating any open-source AI project.

Enterprise Market

Enablement

Integrated safety detection and compliance testing tools help enterprises meet EU AI Act requirements before market entry.

Open-Source

Ecosystem Growth

A credible evaluation framework attracts global developers and researchers to contribute benchmarks, datasets, and methods.



1

Phase 1

Now – 2026 H1

Foundation

- Core framework migration to Eclipse Foundation
- Initial evaluation task set + baseline reports
- First committers confirmed, governance established
- Documentation and onboarding for contributors

2

Phase 2

2026 H2

Capability

- Arena internationalization — multilingual preference
- VLM & Embodied evaluation expansion
- Agent evaluation framework
- Compliance toolkit for GPAI providers

3

Phase 3

2027+

Global Alliance

- AI application system evaluation (beyond models)
- Regional evaluation chapters worldwide
- Behavioral analysis & real-world usage bridging
- Continuous benchmark refresh to combat saturation



Get Involved — Build Trust in AI Together

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Contribute

Propose benchmarks, datasets, and metrics on the project repo

Join

Eclipse PanEval Working Group
Strategic / Contributing / Committer

Adopt

Use PanEval for evaluation, compliance testing & reliability

Resources & Links

GitHub	https://github.com/eclipse-paneval
Maillist	paneval-dev@eclipse-foundation.org
Platform	https://flageval.baai.ac.cn
Website	https://eclipse.dev/paneval

License: Apache 2.0 · Contact: qg.zhu@baai.ac.cn



Benchmarking trust is a community job.

Five years of building.
Now let's build it together — openly.

Qigang Zhu

Beijing Academy of Artificial Intelligence (BAAI)

